

Age-Controlled, Speaker-Independent Voice Pathology Screening with Multifractal and Acoustic Features on the German Saarbrücken Voice Database

Aksaya Kanthan K S¹, Kiruthik Nandhan Murthi¹[0009–0004–8696–3752],
Manasa V¹, Rithvik Rajesh¹, and Jyothish Lal G¹[0000–0001–8925–4776]

Amrita School of Artificial Intelligence, Coimbatore, Amrita Vishwa Vidyapeetham,
India.

cb.ai.u4aid23003@cb.students.amrita.edu,
cb.ai.u4aid23018@cb.students.amrita.edu,
cb.ai.u4aid23022@cb.students.amrita.edu,
cb.ai.u4aid23032@cb.students.amrita.edu, g_jyothishlal@cb.amrita.edu

Abstract. Voice pathology classification is difficult because pathological speech is nonlinear and varies across time scales, and traditional acoustic metrics describe only part of that behaviour. This paper presents a method for both binary (healthy/pathological) and multi-class (neurological/structural disease group) classification of vocal pathology. It combines multifractal features extracted with Multifractal Detrended Fluctuation Analysis (MF DFA) and standardised acoustic features from the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPS). Experiments on the German Saarbrücken Voice Database use strict speaker-independent cross-validation to rule out speaker-level leakage, a control that much of the earlier literature omits. With age and sex as inputs, the method reaches a macro F1 of 0.882 and a balanced accuracy of 0.882 on binary pathology classification, and a macro F1 of 0.649 on disease-group classification, both under five-fold speaker-independent cross-validation. Healthy speakers in this corpus are much younger than pathological ones; without age and sex the features reach 0.760, and 0.679 on age- and sex-matched subsets. A feature ablation run under the same protocol puts the MF DFA descriptors alone at a macro F1 of 0.644, against 0.766 when they are combined with eGeMAPS, which alone reaches 0.765, so the scaling behaviour of the signal carries pathology information but adds little to the standard acoustic parameters.

Keywords: Voice Pathology Detection · Multifractal Analysis · Fractal Dimension · MF-DFA · Saarbrücken Voice Database · Nonlinear Speech Analysis.

1 Introduction

Voice disorders are common. They arise from structural pathology of the vocal folds, from neurogenic disorders of motor coordination, and from functional or

psychogenic causes. Diagnosis normally combines instrumental procedures such as laryngoscopy with perceptual evaluation by a clinician. That combination is accurate, but perceptual judgement is subjective and demands considerable clinical expertise.

Automatic detection is attractive precisely because it removes that subjectivity. Most systems for recognising pathological voices rely on standard spectral parameters such as MFCCs and F0. These describe the spectral properties of speech well, but they capture little of the temporal and nonlinear behaviour typical of pathological voices.

Speech production is a nonlinear process in which airstream initiation, vocal fold vibration, and articulation all interact. In pathological voices these dynamics can become unstable. Fractal and multifractal theory offers a way to quantify that instability.

This work applies multifractal detrended fluctuation analysis to speech recordings from speakers with diagnosed disorders. Multifractal analysis describes a time series through its scaling properties across time scales. A monofractal signal is characterised by a single scaling exponent; a multifractal one needs a spectrum of local exponents, obtained by varying the moment order q of the analysis.

The question addressed here is whether the multifractal descriptors obtained through MFDFA help detect voice pathology, either on their own or alongside standard acoustic parameters.

This work differs from earlier fractal-based voice pathology studies in four ways. First, we use the full multifractal spectrum from MFDFA rather than a single scalar fractal dimension, which gives scaling information across several moment orders. Second, one pipeline handles both binary (healthy vs. pathological) and multi-class (neurological vs. structural) classification, where most earlier studies treat only binary detection. Third, cross-validation is strictly speaker-independent (`StratifiedGroupKFold`), so recordings from one speaker never appear in both the training and the test split. This guards against the data leakage known to produce optimistic results in much of the earlier literature [17]. Fourth, because healthy speakers in this corpus are much younger than pathological ones, results are also reported without age and sex as inputs, next to age-only and sex-only baselines, and on age- and sex-matched subsets. A feature ablation study then quantifies what MFDFA contributes alone and in combination for each classification task.

2 Related Work

Automatic detection of voice pathologies has traditionally relied on acoustic features such as MFCCs, perturbation measures, and spectral statistics combined with statistical or machine learning classifiers [5,1,15]. Standardised clinical frameworks have further formalised multi-parameter acoustic assessment: the Dysphonia Severity Index (DSI) combines jitter, phonation time, and intensity into a perceptually anchored score [20], while standardised assessment proto-

cols specify perturbation and noise measures as a clinical evaluation battery [3]. Although these spectral and perturbation-based descriptors capture important phonatory characteristics, they are inherently linear and cannot represent the nonlinear or multi-scale temporal irregularities often present in pathological voices.

Nonlinear analysis methods have therefore also been explored for pathological speech. Little et al. demonstrated that recurrence quantification analysis and fractal scaling properties extracted from sustained phonations can complement conventional acoustic measures for voice disorder detection, achieving correct classification performance above 91% [13]. Nonlinear dynamical measures such as correlation dimension, Lyapunov exponents, and entropy statistics have been applied to separate healthy from pathological voices with high accuracy on clinical voice databases [6,12]. Temporal complexity measures therefore carry information that standard spectral descriptors miss. Recurrence plot-based characterisation has also been applied to specific vocal fold pathologies including hyperfunctional dysphonia, laryngitis, and vocal fold polyp, achieving classification accuracy above 90% on the Saarbrücken Voice Database [9].

Beyond binary detection, recent work has addressed multi-class voice disorder classification. Orozco-Arroyave et al. found that different feature families are required to capture the distinct acoustic signatures of neurological, functional, and laryngeal disorders [14]. Fractal dimension in multiresolution frameworks has also achieved detection rates above 90% via per-subband power-spectrum analysis [2,10].

Multifractal detrended fluctuation analysis (MFDFA) extends scalar fractal characterisation by modelling scaling behaviour across multiple moments and time scales [8]. Multifractal methods have proven effective in other biomedical domains, most notably in demonstrating the multifractal structure of healthy versus diseased heartbeat dynamics [7]. Closer to the present work, multifractal analysis has been applied directly to speech signals for glottal activity detection, which shows that the method can capture nonlinear phonatory dynamics [11]. Speech-based disease detection has also benefited from careful acoustic feature design: Tsanas et al. developed 132 dysphonia measures from sustained vowels for high-accuracy Parkinson’s disease classification, and showed that engineered descriptors can match or exceed purely data-driven methods in clinical tasks [19]. Sai Jahnvi et al. likewise detected Parkinson’s disease from speech alone using excitation source, voice quality, and prosodic features from sustained vowels [18]. MFDFA applied to voice pathology detection, and particularly to multi-class classification, remains comparatively underexplored.

One methodological concern cuts across all of this work: speaker-level data leakage. Sáenz-Lechón et al. found that systems evaluated without controlling for speaker identity yield optimistic estimates that fail to generalise to unseen speakers, and recommended speaker-independent cross-validation as standard practice [17].

The review points to three gaps. First, fractal work on voice pathology mostly relies on single scalar dimensions rather than the full multifractal spectrum, and

focuses on binary detection. Second, speaker-independent evaluation, although recommended, is omitted in much of the earlier work. Third, healthy and pathological groups in clinical corpora can differ in age and sex, which can inflate results in the same way as speaker leakage. This work addresses all three.

3 Dataset Description

The experiments in this study use recordings from the Saarbrücken Voice Database (SVD) [16], a widely used clinical corpus of healthy and pathological voices. The original audio is stored in the proprietary `.nsp` format and is converted to standard `.wav` for preprocessing and feature extraction. SVD contains several vowels and connected-speech recordings, but this work uses only the sustained vowel `/a/` at normal pitch (`a_n`), which gives a stable phonation condition for cross-speaker comparison. Table 1 summarises the dataset used in the experiments.

Table 1. Dataset Statistics

Statistic	Value
Total recordings (all tokens)	23,119
Unique speakers	1,355
Pathology classes	12 (+ healthy)
Samples for token (<code>a_n</code>)	1,656
Sex distribution	65% female, 35% male
Mean age, healthy / pathological ^a	27.1 / 54.0 years

^aIn the classification set of Section 5.

The `a_n` subset is class-imbalanced, as clinical speech datasets usually are. Pathology classes with too few speakers for reliable speaker-independent cross-validation are excluded. Parkinson’s disease has a single `a_n` recording and hypotonic dysphonia five, so neither is analysed on its own. The nine disorders retained span neurological, structural, and functional aetiologies: recurrent laryngeal nerve paralysis (breathiness from vocal fold immobility); spasmodic dysphonia (involuntary voice breaks from laryngeal dystonia); vocal fold nodules (hoarseness from bilateral overuse lesions); vocal fold polyp (breathiness from asymmetric fold vibration); Reinke’s edema (a lowered, hoarse voice from swelling of the vocal fold cover); laryngitis (hoarseness from laryngeal mucosal inflammation); hyperfunctional dysphonia (strained voice from excessive laryngeal tension); psychogenic dysphonia (voice disorder of psychological aetiology without organic pathology); and functional dysphonia (dysphonia from maladaptive laryngeal behaviour without an identifiable organic cause).

The classes also differ in age: in the classification set (Section 5), healthy speakers are 27.1 ± 10.9 years old on average and pathological speakers 54.0 ± 15.0 ,

and only 100 of SVD’s 687 healthy `a_n` recordings come from speakers over 40. Section 6 controls for this.

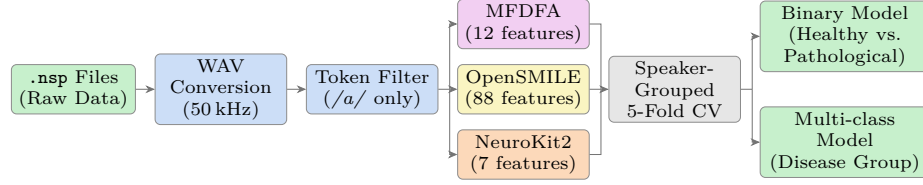


Fig. 1. Overview of the preprocessing and feature extraction pipeline. Raw NSP recordings are converted to WAV format, metadata manifests are generated, and sustained vowel tokens are selected. Multiple feature families are then extracted and merged before training classification models using speaker-grouped cross-validation.

4 Feature Extraction

Each recording is decoded to mono at its native 50 kHz sampling rate and amplitude-normalised before any feature is computed. Three signal-derived families are then extracted, giving 107 features in total: 12 multifractal descriptors, 88 standardised paralinguistic parameters, and 7 nonlinear complexity measures. Two speaker metadata variables are appended to these.

4.1 Multifractal Features

Multifractal detrended fluctuation analysis quantifies how the fluctuations of a signal scale with the size of the window over which they are measured, and how that scaling changes when large and small fluctuations are weighted differently [8]. For a recording $x(k)$ of length N , the analysis starts from the cumulative profile

$$Y(i) = \sum_{k=1}^i [x(k) - \bar{x}], \quad i = 1, \dots, N, \quad (1)$$

which is divided into non-overlapping windows of length s from both ends of the series. A polynomial of order m is fitted within each window v and removed, leaving the local variance $F^2(v, s)$. Averaging these variances with moment order q gives the fluctuation function

$$F_q(s) = \left\{ \frac{1}{2N_s} \sum_{v=1}^{2N_s} [F^2(v, s)]^{q/2} \right\}^{1/q}, \quad (2)$$

where $N_s = \lfloor N/s \rfloor$. Positive q emphasises windows with large fluctuations, negative q those with small ones. The case $q = 0$ requires a separate logarithmic form

and is excluded here. When the signal is self-similar, $F_q(s)$ follows a power law $F_q(s) \sim s^{h(q)}$, and the generalised Hurst exponent $h(q)$ is obtained as the slope of a least-squares fit of $\log F_q(s)$ against $\log s$. A signal is monofractal when $h(q)$ is constant and multifractal when it varies with q .

The exponent is converted to the classical Rényi exponent and then, by Legendre transform, to the singularity spectrum:

$$\tau(q) = q h(q) - 1, \quad \alpha = \frac{d\tau(q)}{dq}, \quad f(\alpha) = q \alpha - \tau(q), \quad (3)$$

with the derivative evaluated numerically over the q grid. The spectrum $f(\alpha)$ describes the fractal dimension of the subset of the signal that carries local exponent α . Its width

$$\Delta\alpha = \alpha_{\max} - \alpha_{\min} \quad (4)$$

measures the strength of the multifractality: a monofractal signal collapses to a point, while a signal whose local regularity varies from instant to instant produces a broad spectrum. The balance between its two arms is summarised by

$$A = \frac{\alpha_{\text{peak}} - \alpha_{\min}}{\alpha_{\max} - \alpha_{\text{peak}}}, \quad (5)$$

where α_{peak} is the exponent at which $f(\alpha)$ is maximal. Values above one indicate a longer left arm, so the scaling is governed mainly by the large-amplitude fluctuations; values below one point to the small-amplitude ones. Both quantities are relevant to pathological phonation, where irregular vocal fold vibration may broaden the spectrum and shift the weight between the two arms relative to healthy speech, in the same way that multifractal structure separates healthy from diseased dynamics in other physiological signals [7]. Twelve descriptors summarise this analysis for each recording; Table 2 lists them.

The configuration used for all reported experiments applies first-order detrending ($m = 1$), which removes a linear trend from each window and leaves the fluctuations around it. The moment orders run from -5 to 5 in steps of 0.5 , so 21 values are requested and 20 remain after $q = 0$ is dropped. Window sizes are 40 logarithmically spaced integer scales between 16 samples (0.32 ms at 50 kHz) and $N/4$, which for the median token length of 1.27 s corresponds to roughly 320 ms. The lower bound keeps enough points per window for the local fit, and the upper bound keeps enough windows per scale for the average in Eq. (2) to be stable. A recording is only accepted when at least six distinct scales are available and each $h(q)$ fit has at least three finite points; all 1,656 `a_n` recordings met these conditions.

4.2 OpenSMILE Features

Standardised acoustic features are extracted with the OpenSMILE toolkit [4] using the extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPSv02), a set designed for reproducible comparison across paralinguistic studies. It applies

Table 2. MFDFA Descriptors Extracted per Recording

Descriptor	Interpretation
$h(q)$ mean, std	Average scaling exponent and its variation across moment orders; the spread is a first indicator of multifractality
$h(q)$ min, max	Scaling of the smallest and largest fluctuations
$\tau(q)$ mean, std	Location and curvature of the Rényi exponent; a strictly linear $\tau(q)$ implies monofractal scaling
α mean, std	Centre and spread of the local Hölder exponents
$\Delta\alpha$	Spectrum width, i.e. the strength of the multifractality
$\alpha_{\text{peak}}, f(\alpha_{\text{peak}})$	Position and height of the spectrum maximum, i.e. the dominant scaling regime
A	Left/right asymmetry of the spectrum

functionals to 25 low-level descriptors and yields 88 parameters. The frequency-related descriptors cover fundamental frequency in the semitone scale, local jitter, and the frequency, bandwidth, and relative amplitude of the first three formants. The energy and amplitude group covers loudness, local shimmer, and the harmonics-to-noise ratio. The spectral group covers the alpha ratio, the Hammarberg index, spectral slopes in the 0–500 Hz and 500–1500 Hz bands, the relative energies H1–H2 and H1–A3, spectral flux, and the first four MFCCs. Temporal descriptors cover the rate of loudness peaks and the length and rate of voiced and unvoiced segments. The functionals are the arithmetic mean and the normalised standard deviation over the relevant frames, with percentiles, ranges, and rising and falling slopes added for pitch and loudness. The set includes the perturbation measures used in clinical practice, namely jitter, shimmer, and the harmonics-to-noise ratio, so the ablation in Section 6 measures the multifractal descriptors against an established acoustic baseline. It also covers properties such as formant structure and spectral tilt, which scaling analysis does not represent.

4.3 NeuroKit2 Features

Seven further features characterise the nonlinear dynamics of the waveform from a different direction than MFDFA. Three are entropy measures: Shannon entropy in bits, approximate entropy, and sample entropy, the latter two with embedding dimension 2 and a tolerance of 0.2 times the standard deviation of the signal. Two are single-valued fractal dimensions computed directly from the waveform, the Petrosian and Sevcik estimators. The remaining two are the skewness and kurtosis of the amplitude distribution. Because approximate and sample entropy are quadratic in the number of samples, the entropy and fractal-dimension measures are computed on the signal decimated to 8 kHz, which keeps the extraction

tractable over the full corpus; skewness and kurtosis are computed at full resolution. These descriptors overlap with the multifractal ones, since both quantify signal irregularity, but they do so with single-scale statistics rather than across a range of window sizes. The ablation in Section 6 shows how much they add once MFDFA is already present.

4.4 Speaker Metadata

Two auxiliary features come from the SVD speaker metadata: age, computed from the interval between the birth date and the recording date, and biological sex. Every configuration is evaluated both with and without these two variables, and each is also tested on its own as a baseline.

5 Data Pipeline and Splitting Strategy

The overall preprocessing workflow is illustrated in Figure 1. The raw `.nsp` recordings are converted to `.wav`, indexed through a metadata manifest, normalised, and then processed by the four feature-extraction modules before the resulting feature tables are merged into a unified machine learning dataset. To minimise variability and within-speaker duplication, most experiments use only the sustained vowel `a_n`.

Speaker-level leakage is the main risk in clinical voice datasets. We therefore use `StratifiedGroupKFold` with speaker identity as the grouping variable, so all recordings from a speaker stay in the same fold. The classification set is built in three steps. Pathological recordings outside the six disorders assigned to the two disease groups are removed (489 recordings of hyperfunctional, functional, psychogenic and hypotonic dysphonia and Reinke’s edema). Speakers with both healthy and pathological recordings are also removed. Finally, the healthy class is randomly downsampled from 685 to 478 recordings to match the pathological count. The result has 956 recordings from 820 speakers: 478 healthy, 277 neurological and 201 structural. The disease-group task uses the 478 pathological recordings.

To control for demographics, we also build age- and sex-matched subsets. Each pathological recording is paired with the healthy recording of the same sex that is closest in age, within two years, drawn without replacement from all healthy recordings; recordings without a match are dropped. This gives 158 pairs (316 recordings). For the disease-group task, structural recordings are matched to neurological ones in the same way (139 pairs). SVD has few healthy speakers over 60, so the matched subsets cover ages of roughly 15 to 75.

All models use the full set of 107 features without feature selection. Logistic regression, random forest, LightGBM and XGBoost are trained for the binary task, and the last three for the disease-group task; the binary decision threshold is tuned on each training fold. We report the mean and standard deviation over the five speaker-grouped folds for the best classifier of each configuration. Differences between classifiers were mostly below 0.05 macro F1, and at most 0.09 (for the sex-only baseline).

6 Experimental Setup and Results

This section evaluates how well the multifractal features discriminate on their own and what they add to conventional acoustic descriptors, with and without age and sex as inputs. All experiments use five-fold speaker-independent cross-validation (`StratifiedGroupKFold`), so no speaker appears in both the training and the test split. Performance is reported as accuracy, balanced accuracy, and macro-averaged F1; the latter two carry more weight here because of the class imbalance. In preliminary experiments, naive random splits produced far higher scores, which is why every result below uses speaker-independent splits.

6.1 Speaker-Independent Evaluation

Only the sustained vowel token `a_n` is used to ensure consistency across speakers. Two classification tasks are considered: (1) binary classification of healthy vs. pathological speakers, and (2) disease-group classification of neurological vs. structural disorders. The binary task is a screening scenario: decide whether a voice disorder is present. Table 3 gives the results for all 107 signal features with and without age and sex, for the demographic variables alone, and on the matched subset.

Table 3. Binary Classification Results (Speaker-Grouped CV, Mean $\pm$ SD over Five Folds)

Inputs	Set	Best Model	Bal. Acc.	F1 Macro
Age only	Main	LogReg	0.832	0.830 ± 0.072
Sex only	Main	XGBoost	0.538	0.532 ± 0.055
Age + sex	Main	XGBoost	0.846	0.846 ± 0.017
Signal features	Main	Random Forest	0.762	0.760 ± 0.042
Signal features + age + sex	Main	XGBoost	0.882	0.882 ± 0.025
Age + sex	Matched	LogReg	0.447	0.386 ± 0.061
Signal features	Matched	LightGBM	0.683	0.679 ± 0.044

With age and sex among the inputs, the best model reaches a macro F1 of 0.882, but age alone already reaches 0.830, while the signal features without demographic inputs reach 0.760. On the matched subset, the age and sex baseline falls to chance and the signal features still reach 0.679. Part of this drop comes from the smaller training set: ten random subsets of the main set with the same size (316 recordings) reach 0.734 ± 0.023 . These numbers sit well below those from the random-split runs, which is the expected cost of a more realistic protocol. Figure 2 (top) shows the confusion matrices of the models trained on signal features alone.

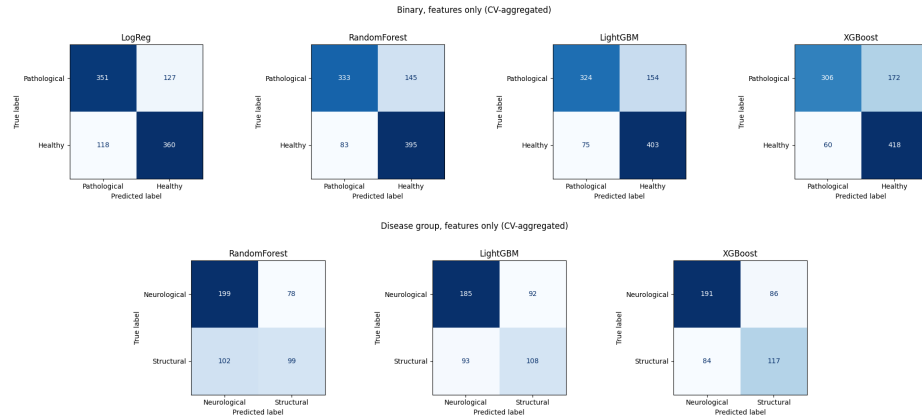


Fig. 2. Confusion matrices with signal features only, aggregated over the five folds, one panel per classifier. Top: binary classification (main set). Bottom: disease-group classification.

6.2 Disease Group Classification

Detecting pathology is one task. Sorting disorders by clinical cause is a harder one. The pathologies here are split into two clinical categories. The neurological group contains recurrent laryngeal nerve paralysis (212 recordings), spasmodic dysphonia (64) and the single Parkinson’s disease recording; the structural group contains laryngitis (140), vocal fold polyp (44) and vocal fold nodules (17). Neurological disorders come from abnormal neural control of the vocal folds, structural disorders from abnormalities in the tissue of the folds themselves. Separating the two groups from the speech signal alone is difficult. Table 4 summarises the performance of the evaluated models for disease-group classification.

Table 4. Disease Group Classification Results (Mean $\pm$ SD over Five Folds)

Inputs	Set	Best Model	Bal. Acc.	F1 Macro
Age only	Main	Random Forest	0.553	0.536 ± 0.069
Sex only	Main	XGBoost	0.606	0.605 ± 0.091
Age + sex	Main	XGBoost	0.604	0.592 ± 0.083
Signal features	Main	XGBoost	0.636	0.635 ± 0.058
Signal features + age + sex	Main	XGBoost	0.652	0.649 ± 0.066
Age + sex	Matched	LightGBM	0.341	0.332 ± 0.058
Signal features	Matched	Random Forest	0.580	0.575 ± 0.042

Disease-group classification is much harder than binary pathology detection. The best model reaches an F1 of 0.635 without demographic inputs and 0.649

with them, so age plays a smaller part here than in the binary task. On the matched subset the score falls to 0.575, a drop that random subsets of the same size do not show (0.643 ± 0.022). The acoustic differences between neurological and structural disorders are far less clear-cut than those between healthy and pathological voices. Figure 2 (bottom) shows the confusion matrices for this task.

6.3 Feature Ablation Analysis

A feature ablation study measures what each feature family contributes. Combinations of multifractal features, OpenSMILE descriptors, and NeuroKit2 features were evaluated under the same speaker-grouped cross-validation protocol, with and without age and sex, and on the matched subsets. Table 5 lists the results together with the demographic baselines.

Table 5. Feature Ablation (Macro F1, Mean $\pm$ SD over Five Folds, Best Classifier per Cell). Signal features are used without age and sex except in the “+ age, sex” column.

Features	Binary			Disease group	
	Main	Main + age, sex	Matched	Main	Matched
MF DFA	0.644 ± 0.027	0.854 ± 0.015	0.612 ± 0.028	0.560 ± 0.079	0.540 ± 0.115
OpenSMILE	0.765 ± 0.036	0.884 ± 0.024	0.662 ± 0.063	0.624 ± 0.067	0.580 ± 0.040
NeuroKit2	0.687 ± 0.036	–	0.610 ± 0.051	0.559 ± 0.059	0.585 ± 0.084
MF DFA + OpenSMILE	0.766 ± 0.039	0.882 ± 0.028	0.664 ± 0.022	0.621 ± 0.080	0.587 ± 0.046
MF DFA + NeuroKit2	0.691 ± 0.027	0.868 ± 0.031	0.627 ± 0.041	0.566 ± 0.091	0.562 ± 0.061
OpenSMILE + NeuroKit2	0.767 ± 0.051	0.882 ± 0.025	0.668 ± 0.049	0.626 ± 0.078	0.591 ± 0.046
All three	0.760 ± 0.042	0.882 ± 0.025	0.679 ± 0.044	0.635 ± 0.058	0.575 ± 0.042
Age only	0.830 ± 0.072		0.451 ± 0.040	0.536 ± 0.069	0.420 ± 0.066
Sex only	0.532 ± 0.055		0.435 ± 0.065	0.605 ± 0.091	0.429 ± 0.052
Age + sex	0.846 ± 0.017		0.386 ± 0.061	0.592 ± 0.083	0.332 ± 0.058

Without age and sex, the multifractal features on their own reach an F1 of 0.644 (0.612 on the matched subset), so the scaling behaviour of the signal carries some information about pathology. OpenSMILE alone reaches 0.765, and adding MF DFA to it changes the score by less than 0.02, so the two families are largely redundant. The same holds for disease groups, where MF DFA alone reaches 0.560.

Adding NeuroKit2 changes almost nothing, so the complexity information it measures is largely already captured by the other descriptors.

Age has a much larger effect than any of that. Age and sex alone reach 0.846, and adding them to any feature set raises the binary score to between 0.854 and 0.884, while on the matched subsets they fall to chance. Figure 3 compares all configurations.

6.4 Per-Disease Binary Detection

A separate binary classifier was trained for each disease against the healthy class to see which pathologies are individually detectable, with and without age and

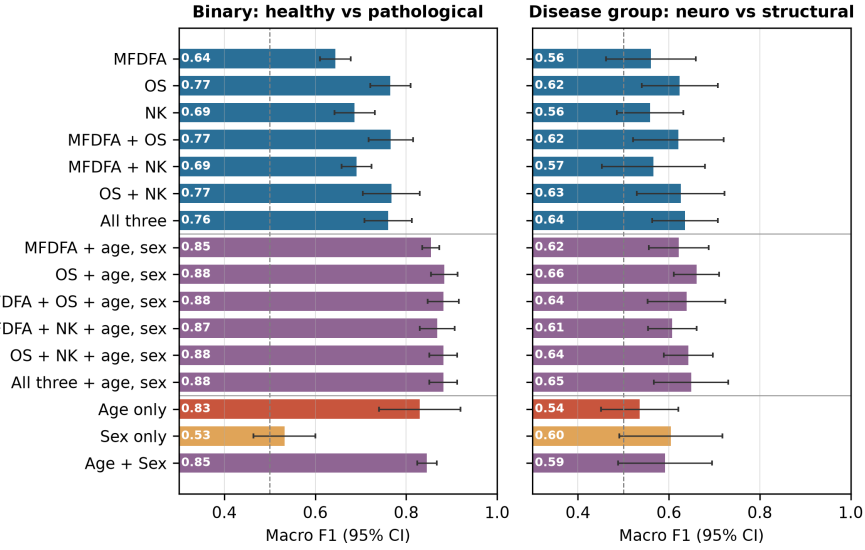


Fig. 3. Feature ablation on the main set with 95% confidence intervals over the five folds. Left: binary classification. Right: disease-group classification. OS: OpenSMILE, NK: NeuroKit2. Bar colour marks the inputs: signal features only (blue), with age and sex (purple), age or sex alone (red, orange).

sex, and on age- and sex-matched subsets (omitted when fewer than 20 pairs are found).

Table 6. Per-Disease Binary Detection (Macro F1, Mean $\pm$ SD over Five Folds, Best Classifier per Cell)

Disease	n	Features + age, sex	Features only	Matched (pairs)
Recurrent laryngeal nerve paralysis	212	0.882 $\pm$ 0.028	0.786 $\pm$ 0.032	0.696 $\pm$ 0.085 (85)
Reinke’s edema	67	0.828 $\pm$ 0.062	0.777 $\pm$ 0.056	0.742 $\pm$ 0.051 (45)
Psychogenic dysphonia	88	0.806 $\pm$ 0.072	0.714 $\pm$ 0.140	0.633 $\pm$ 0.116 (59)
Laryngitis	140	0.772 $\pm$ 0.040	0.683 $\pm$ 0.065	0.601 $\pm$ 0.119 (99)
Functional dysphonia	107	0.766 $\pm$ 0.044	0.610 $\pm$ 0.052	0.562 $\pm$ 0.087 (85)
Vocal fold polyp	44	0.754 $\pm$ 0.141	0.683 $\pm$ 0.068	0.544 $\pm$ 0.164 (36)
Hyperfunctional dysphonia	205	0.753 $\pm$ 0.067	0.657 $\pm$ 0.030	0.602 $\pm$ 0.080 (132)
Spasmodic dysphonia	64	0.699 $\pm$ 0.184	0.663 $\pm$ 0.226	0.713 $\pm$ 0.227 (25)
Vocal fold nodules	17	0.683 $\pm$ 0.169	0.683 $\pm$ 0.169	– (16)

Recurrent laryngeal nerve paralysis and Reinke’s edema are the most acoustically distinct and remain detectable on matched data (0.696 and 0.742). Most other disorders lose accuracy once age is removed, and vocal fold polyp (0.544) and functional dysphonia (0.562) fall close to chance. Spasmodic dysphonia (12 speakers) and vocal fold nodules (17 recordings) are too small for reliable esti-

mates. Screening for pathology in general is feasible; identifying which pathology is not, and closing that gap will likely need larger and more specialised datasets.

7 Conclusion

This paper investigates multifractal descriptors obtained via MFDFA for automated voice pathology detection on the Saarbrücken Voice Database. MFDFA features are combined with conventional acoustic (OpenSMILE) and complexity-based (NeuroKit2) descriptors and evaluated under strict speaker-independent cross-validation, so the reported scores apply to unseen speakers. Because healthy speakers in this corpus are much younger than pathological ones, results are also reported without age and sex as inputs and on age- and sex-matched subsets.

With age and sex as inputs, binary pathology detection reaches a macro F1 of 0.882; without them the features reach 0.760, and 0.679 on the matched subsets. The ablation shows that the MFDFA descriptors alone reach 0.644 without any conventional acoustic features, but add little once these are present. Disease-group classification is considerably weaker at 0.649 macro F1 (0.575 on matched subsets), and per-disease detection on matched subsets ranges from 0.742 down to 0.544 F1. For now the method suits screening better than differential diagnosis.

This study is limited to a single sustained vowel, and the matched evaluation covers speakers of roughly 15 to 75 years, since the corpus contains few healthy speakers over 60. Matching also favours younger patients, so part of the lower matched scores may reflect milder disease. Next steps are validation on larger independent cohorts with age-balanced controls, extension to multiple tokens and to end-to-end models, and an analysis of which parts of the multifractal spectrum carry the clinically useful signal.

References

1. Ahmed Al-Nasheri, Ghulam Muhammad, and Mansour Alsulaiman. Voice pathology detection using MFCC and support vector machines. *Journal of Voice*, 31(4):512.e1–512.e8, 2017.
2. Zulfiqar Ali, Irraivan Elamvazuthi, Mansour Alsulaiman, and Ghulam Muhammad. Detection of voice pathology using fractal dimension in a multiresolution analysis of normal and disordered speech signals. *Journal of Medical Systems*, 40, 2016.
3. Philippe H. Dejonckere, Patrick Bradley, Pedro Clemente, Guy Cornut, Lise Crevier-Buchman, Gerhard Friedrich, Paul Van de Heyning, Marc Remacle, and Virginie Woisard. A basic protocol for functional assessment of voice pathology, especially for investigating the efficacy of (phonosurgical) treatments and evaluating new assessment techniques. *European Archives of Oto-Rhino-Laryngology*, 258(2):77–82, 2001.
4. Florian Eyben, Klaus R. Scherer, Björn W. Schuller, Johan Sundberg, Elisabeth André, Carlos Busso, Laurence Y. Devillers, Julien Epps, Petri Laukka,

- Shrikanth S. Narayanan, and Khiet P. Truong. The Geneva minimalistic acoustic parameter set (GeMAPS) for voice research and affective computing. *IEEE Transactions on Affective Computing*, 7(2):190–202, 2016.
5. Juan I. Godino-Llorente, Pedro Gómez-Vilda, and Manuel Blanco-Velasco. Dimensionality reduction of a pathological voice quality assessment system based on Gaussian mixture models and short-term cepstral parameters. *IEEE Transactions on Biomedical Engineering*, 53(10):1943–1953, 2006.
6. Patricia Henríquez, Jesús B. Alonso, Miguel A. Ferrer, Carlos M. Travieso, Juan I. Godino-Llorente, and Fernando Díaz-de María. Characterization of healthy and pathological voice through measures based on nonlinear dynamics. *IEEE Transactions on Audio, Speech, and Language Processing*, 17(6):1186–1195, 2009.
7. Plamen Ch. Ivanov, Luís A. N. Amaral, Ary L. Goldberger, Shlomo Havlin, Michael G. Rosenblum, H. Eugene Stanley, and Zbigniew R. Struzik. Multifractality in human heartbeat dynamics. *Nature*, 399:461–465, 1999.
8. Jan W. Kantelhardt, Stephan A. Zschiegner, Eva Koscielny-Bunde, Shlomo Havlin, Armin Bunde, and H. Eugene Stanley. Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A: Statistical Mechanics and its Applications*, 316:87–114, 2002.
9. A. G. Krishnan, G. Jyothish Lal, and E. A. Gopalakrishnan. A novel and accurate recurrence plot-based characterization of speech dynamics in hyperfunctional dysphonia, laryngitis, and vocal fold polyp. *IEEE Access*, 14, 2026.
10. Rakesh Kumar and Pranab Mullik. Fractal dimension based voice pathology detection. *Applied Acoustics*, 76:248–252, 2014.
11. G. Jyothish Lal, E. A. Gopalakrishnan, and D. Govind. Glottal activity detection from the speech signal using multifractal analysis. *Circuits, Systems, and Signal Processing*, 39(4):2118–2150, 2020.
12. Max A. Little, Patrick E. McSharry, Eric J. Hunter, Jennifer Spielman, and Lorraine O. Ramig. Suitability of dysphonia measurements for telemonitoring of Parkinson’s disease. *IEEE Transactions on Biomedical Engineering*, 56(4):1015–1022, 2009.
13. Max A. Little, Patrick E. McSharry, Stephen J. Roberts, Declan A. E. Costello, and Irene M. Moroz. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *BioMedical Engineering OnLine*, 6:23, 2007.
14. Juan R. Orozco-Arroyave, Julian D. Arias-Londoño, Jesús F. Vargas-Bonilla, María C. González-Rativa, and Elmar Nöth. Characterization methods for the detection of multiple voice disorders: Neurological, functional, and laryngeal diseases. *IEEE Journal of Biomedical and Health Informatics*, 19(6):1820–1828, 2015.
15. S. Rai, S. Rishita, K. S. Lasya, and Susmitha Vekkot. Design and development of an e-health package for voice pathology detection. In *Proceedings of Springer LNNS*, 2025.
16. Saarbrücken Voice Database. <http://stimmdb.coli.uni-saarland.de/>. last accessed 2026/07/27.
17. Nicolás Sáenz-Lechón, Juan I. Godino-Llorente, Víctor Osma-Ruiz, and Pedro Gómez-Vilda. Methodological issues in the development of automatic systems for voice pathology detection. *Biomedical Signal Processing and Control*, 1(2):120–128, 2006.
18. B. Sai Jahnavi, B. Sai Supraja, and S. Lalitha. A vital neurodegenerative disorder detection using speech cues. *Journal of Intelligent & Fuzzy Systems*, 38(5):6337–6345, 2020.

19. Athanasios Tsanas, Max A. Little, Patrick E. McSharry, Jennifer Spielman, and Lorraine O. Ramig. Novel speech signal processing algorithms for high-accuracy classification of Parkinson's disease. *IEEE Transactions on Biomedical Engineering*, 59(5):1264–1271, 2012.
20. Floris L. Wuyts, Marc S. De Bodt, Geert Molenberghs, Marc Remacle, Louis Heylen, Bertrand Millet, Kristiane Van Lierde, Jos Raes, and Paul H. Van de Heyning. The Dysphonia Severity Index: An objective measure of vocal quality based on a multiparameter approach. *Journal of Speech, Language, and Hearing Research*, 43(3):796–809, 2000.